

8 Analytic Geometry and Calculus

8.1 Axes and Co-ordinates

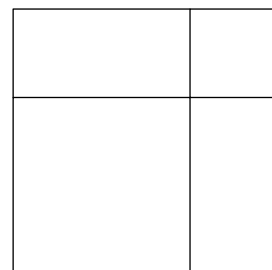
Modern mathematics is dominated by *algebra*. We trust equations and the rules of algebra more than pictorial, geometric reasoning. For example, when modern mathematicians are likely to interpret the expression $(x + y)^2 = x^2 + 2xy + y^2$ within the laws (axioms) of algebra:

$$\begin{aligned}
 (x + y)^2 &= (x + y)(x + y) && \text{(definition of 'square')} \\
 &= x(x + y) + y(x + y) && \text{(distributive law)} \\
 &= x^2 + xy + yx + y^2 && \text{(distributive law twice more)} \\
 &= x^2 + 2xy + y^2 && \text{(commutativity)}
 \end{aligned}$$

For most of mathematical history, however, this result would have been stated *geometrically*, as in Euclid's *Elements* (Thm II. 4):

The square on two parts equals the squares on each part plus twice the rectangle on the parts.

The proof was a simple picture.



We've seen how algebra and algebraic notation were slowly adopted in renaissance Europe. While its utility for efficient calculation was noted, algebra was not initially considered acceptable for the purposes of *proof*; practitioners still felt the need to justify calculations geometrically. In modern times this seems backwards: if a college student today were asked to prove Euclid's result,⁵⁹ they'd likely start by labeling the 'parts' x and y , before appealing to the algebraic formula at the top of the page! Of course, each piece of algebra has a geometric basis:

- Distributivity is the fact that a rectangle on a side and two parts equals the sum of the rectangles on the side and each of the parts respectively.
- Commutativity is the fact that a rectangle has the same area after a 90° rotation.

Modern mathematics has converted geometric rules into algebra and largely forgotten their origin. The slow triumph of algebra over geometry is one of the great revolutions of mathematical history, completely changing the way mathematicians *think*. More practically, algebra permits easy generalization: how would one geometrically justify an expression such as

$$(x + y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4 ?$$

Euclidean Geometry is *synthetic*: based on purely geometric axioms without formulæ or co-ordinates. *Analytic geometry* married algebra, geometry, and number, by introducing *axes* and *co-ordinates*. Modern geometry is primarily analytic and it is now rare to find a mathematician working synthetically—algebra's domination of Euclidean geometry is total. The critical contribution to this revolution was made almost simultaneously by two Frenchmen...

⁵⁹But importantly *not* a 5th-grader, who'd likely consider algebra to be alien and pointless: *isn't the picture obvious?* Keep this in mind if you ever have the privilege of teaching at grade-school: it is *your* ideas that are strange...

Pierre de Fermat (1601–1665)

Fermat made great strides in several areas (probability, analytic geometry, early calculus, number theory, optics), even though mathematics was a pastime rather than his profession.⁶⁰ You've likely encountered his name attached to two famous results in number theory:

Fermat's Little Theorem If p is prime and x an integer, then $x^p \equiv x \pmod{p}$.

Fermat's Last Theorem If $n \in \mathbb{N}_{\geq 3}$, then $x^n + y^n = z^n$ has no integer solutions with $xyz \neq 0$. Fermat is not believed to have proved this beyond a special case ($n = 4$). This problem has captivated mathematicians ever since, motivating several developments in algebra and number theory; a complete proof finally came in the 1990s.

Some of Fermat's fame comes from his enigma. Much of what we know comes from letters to friends in which he claims much but rarely offers complete proofs—it is often unclear whether he merely suspected a general statement. When Fermat died, his notes and letters contained many unproven claims. Leonhard Euler (1707–83) in particular expended much effort on these. Fermat's work on analytic geometry was largely developed by 1629, though it wasn't published during his lifetime. His approach was similar to that of Descartes which we describe below. The two corresponded, not always cordially, but both acknowledged the other's contributions. We'll return to Fermat when we discuss the beginnings of calculus in the next section.

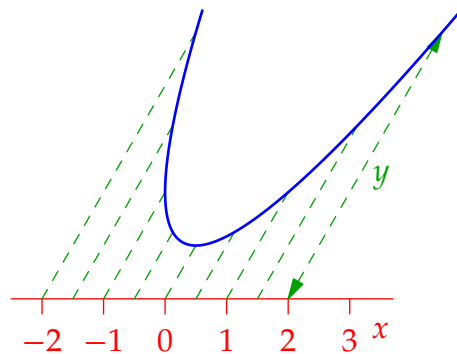
René Descartes (1596–1650)

The chalk to Fermat's cheese, Descartes rigorously recorded everything. His defining work is 1637's *Discours de la méthode*...⁶¹ While enormously influential in philosophy, *Discours* was intended to lay the groundwork for investigation within mathematics and the sciences—Descartes finishes *Discours* by commenting on the necessity of experimentation in science and on his reluctance to publish due to his fear of suffering the same persecution as Galileo.⁶² The copious appendices to *Discours* contain Descartes' scientific work. It is in one of these, *La Géométrie*, that Descartes introduces axes and co-ordinates.

We now think of Cartesian axes and co-ordinates as *plural*, but both Fermat and Descartes used only one axis. Here is a sketch of their approach.

Draw a straight line (the **axis**) containing two fixed points labelled 0 (the *origin*) and 1. All points on the axis are identified with numbers x (originally only positive).

A **curve** is described as an algebraic relationship between x and the **distance** y from the axis to the curve measured using a family of **parallel lines**.



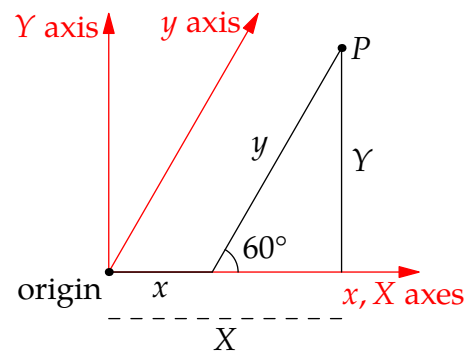
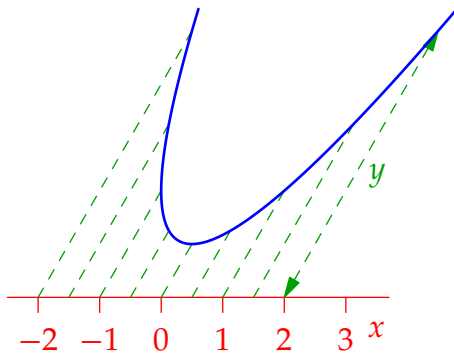
⁶⁰Fermat was wealthy but not aristocratic, attending the University of Orléans for three years where he trained as a lawyer.

⁶¹...of rightly conducting one's reason and of seeking truth in the sciences. The primary part of this work is philosophical and contains his famous phrase *cogito ergo sum* (I think therefore I am).

⁶²At this time, France was still Catholic. Descartes had moved thence to Holland in part to pursue his work more freely. In 1649 Descartes moved to Sweden where he died the following year.

While neither Descartes nor Fermat had a second axis, their approach implicitly imagines one, namely the **measuring line** through the origin. It therefore makes sense for us to speak of the *co-ordinates* (x, y) ; the modern terms *abscissa* (x) and *ordinate* (y) date from shortly after Descartes. It wasn't long before a second axis orthogonal to the first was instituted (Frans van Schooten, 1649), an approach that quickly became standard.

Example 1 The previous picture shows some of the flexibility inherent in Descartes' approach. The curve $y = x^2 + 1$ is drawn, where the 'y-axis' is inclined 60° to the horizontal. To represent the curve in a more standard fashion, we perform a change of co-ordinates. Suppose a point P has co-ordinates (x, y) with respect to the original slanted axes and (X, Y) with respect to the familiar orthogonal axes of van Schooten.



The second picture shows the relationship between the two co-ordinate systems

$$\begin{cases} X = x + y \cos 60^\circ = x + \frac{1}{2}y \\ Y = y \sin 60^\circ = \frac{\sqrt{3}}{2}y \end{cases}$$

A little algebra quickly shows that $\sqrt{3}X - Y = \sqrt{3}x$. It follows that for any point on the curve, we have

$$\begin{aligned} (\sqrt{3}X - Y)^2 &= 3x^2 = 3(y - 1) = 3\left(\frac{2}{\sqrt{3}}Y - 1\right) \\ \Rightarrow 3X^2 - 2\sqrt{3}XY + Y^2 - 2\sqrt{3}Y + 3 &= 0 \end{aligned}$$

This is an implicit equation for a parabola in rectangular co-ordinates. Descartes has essentially chosen a second axis in such a way as to make the equation of this parabola very simple.⁶³

⁶³That the implicit equation is a parabola may be confirmed in modern language by computing its *discriminant*: a non-degenerate quadratic curve $aX^2 + 2bXY + cY^2 + \dots = 0$ is a parabola if and only if $b^2 - ac = 0$. You've possibly encountered this in a linear algebra course, where the standard approach is to *rotate* both axes instead by diagonalizing a symmetric matrix: indeed the discriminant is negative the determinant of the matrix $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ and $(x \ y)A\begin{pmatrix} x \\ y \end{pmatrix} = aX^2 + 2bXY + cY^2 + \dots$

Example 2 Analytic geometry affords easy proofs of many results that are significantly harder in Euclidean geometry. For instance, here is the famous *centroid theorem*.

The three medians⁶⁴ of a triangle meet at a common point (the *centroid*), which moreover lies two-thirds of the way along each median.

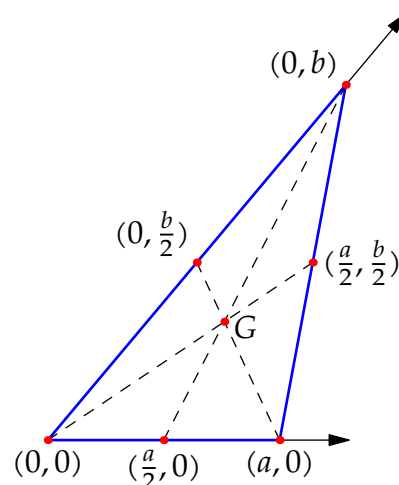
Choose axes pointing along two sides of a triangle with the origin at one vertex. If the two sides lying on the axes have lengths a and b , then the third side necessarily has equation

$$bx + ay = ab$$

Moreover, the midpoints of the three sides plainly have co-ordinates as pictured.

To complete the proof, simply compute the co-ordinates of the point two-thirds of the way along each median: for the median through the upper vertex, this is

$$\frac{2}{3} \left(\frac{a}{2}, 0 \right) + \frac{1}{3} (0, b) = \frac{1}{3} (a, b)$$



One obtains the same co-ordinates for the other two medians, whence *all three meet at a common point G*, the *centroid* of the triangle.

As demonstrated by the centroid proof, Descartes' ability to *choose axes to fit the problem* is a critical advantage of analytic geometry, largely dispensing with the tedious consideration of *congruence* in synthetic geometry.

Descartes found that his method could solve many problems in a more efficient manner than using synthetically; in particular, finding complicated intersections of curves. As we'll see in the next section, such arguments were necessary for the development of calculus, and would often rely on his Factor Theorem (pg. 93).

Given the novelty of his approach, Descartes typically gave geometric proofs to back up his algebraic work. However, he also saw the future: once several examples were done, he claimed that it was no longer necessary to draw physical lines and provide a geometric argument; *the algebra was the proof*.

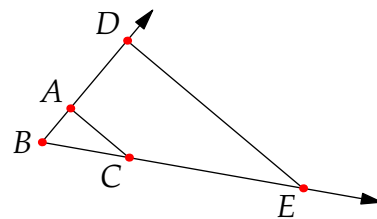
Exercises 8.1. *Key concepts: Basic algebra encodes geometric facts, Analytic geometry helps algebra & number eclipse geometry, Fermat & Descartes' single-axis approach*

1. Assume that $xy = c$ represents a hyperbola with asymptotes the x - and y -axes. Show that $xy + c = rx + sy$ also represents a hyperbola, and find its asymptotes.
2. Determine the locus of the equation $b^2 - 2x^2 = 2xy + y^2$.
(Hint: add x^2 to both sides and remember that axes do not have to be orthogonal...)

⁶⁴A segment joining a vertex and the midpoint of the opposite side.

3. We describe a method whereby Descartes constructed the product of two lengths.

- Let $\overrightarrow{BC}$ and $\overrightarrow{BD}$ be rays forming an acute angle at B . Our goal is to multiply $|BD|$ by $|BC|$.
- Suppose $|AB| = 1$, where A lies on $\overrightarrow{BD}$.
- Join $\overline{AC}$ and draw $\overline{DE}$ parallel to $\overline{AC}$ so that $E \in \overrightarrow{BC}$.

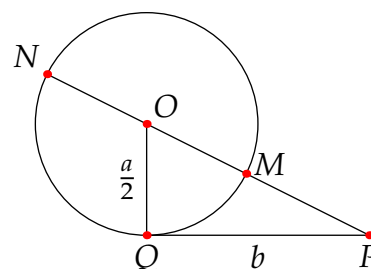


Prove that $|BE|$ is the product of $|BD|$ and $|BC|$.

Similarly, given lengths $|BE|$ and $|BD|$, explain how to construct a segment whose length is the quotient $\frac{|BE|}{|BD|}$.

4. Here is a geometric justification of Descartes for the solution of the equation $x^2 = ax + b^2$, where $a, b > 0$.

- Let $\overline{OQ}$ and $\overline{PQ}$ be perpendicular with lengths $\frac{a}{2}$ and b respectively.
- Draw the circle centered at O with radius $\frac{a}{2}$.
- Draw and extend the line $\overline{OP}$.
- The solution is $x = |NP|$.



- Prove that Descartes' construction indeed recovers a solution.
 - Show that the other solution to the equation $x^2 = ax + b^2$ is negative (*false* to Descartes). How is it visible in the picture?
 - Explain how the same picture could be used to solve the equation $x^2 + ax = b^2$.
5. Following the work of Islamic mathematicians such as Omar Khayyam, Fermat could describe the solutions to certain cubic equations as the intersection of two conics.

For example, to solve $x^3 + bx^2 = bc$ ($b, c > 0$), Fermat would introduce a new variable y by setting both sides of the equation equal to bxy . The (positive) solution to the cubic is the x -solution to the system of equations

$$\begin{cases} x^2 + bx = by \\ xy = c \end{cases}$$

Sketch these curves explicitly for the cubic $x^3 + 4x^2 = 24$. What is the solution?

8.2 The Beginnings of Calculus

At the heart of calculus is the relationship between *velocity*, *displacement*, *rate of change* and *area*.

- The *instantaneous velocity* of a particle is the *rate of change* of its *displacement*. Otherwise said, the *slope* of the displacement-time graph.
- The *displacement* of a particle is the *net area* under its *velocity-time* graph.

These relationships are essentially geometric. As we've seen, the advent of analytic geometry in the 1620-30s facilitated easier geometric computations. With this powerful tool available, the rapid development of calculus was arguably inevitable.

In the context of the above, the *Fundamental Theorem of Calculus* is intuitive: complete knowledge of displacement (from some initial point) is equivalent to complete knowledge of velocity. A modern statement is more daunting.

Theorem (Fundamental Theorem of Calculus).

1. If f is integrable on $[a, b]$, then $F(x) := \int_a^x f(u) \, du$ is continuous on $[a, b]$. Moreover, if f is continuous at $c \in (a, b)$, then $F'(c) = f(c)$.
2. If F is continuous on $[a, b]$ with integrable derivative on (a, b) , then $\int_a^b F'(x) \, dx = F(b) - F(a)$.

The modern version is abstract and very widely applicable: we've gone way beyond considerations of velocity (f) and displacement (F). The challenge of *teaching*⁶⁵ and *proving* the Fundamental Theorem lies in developing and understanding what is meant by *continuous*, *differentiable* and *integrable*. The quest for good definitions of these concepts is the story of analysis in the 17-1800s.

We begin, however, with some older considerations of the velocity and area problems.

The Velocity Problem pre-1600

The modern ideas of *uniform* or *average* velocity are straightforward:

Measure displacement over a given time interval and divide one by the other.

Several ancient Greek mathematicians considered versions of this, and of uniform acceleration, the challenge of considering a ratio of two unlike quantities (displacement : time) proved difficult to surmount. Around 1200, Gerard of Brussels resurrected the ratio approach as a definition of velocity, though it was not considered a numerical quantity in its own right.

Gerard was credited by the Oxford/Merton Thinkers (1330s) as influencing their investigations of *instantaneous velocity*, a much more difficult issue. They offered the following definition.

⁶⁵Calculus students can easily be taught the *mechanics* of calculus (the power law, chain rule, etc.) without having any idea of its *meaning*; witness the power and curse of analytic geometry and algebra!

Definition. The *velocity* of a particle at an instant will be measured as the uniform velocity along the path that would have been taken by the particle if it continued with that velocity.

This is vague, though it teases at the idea of inertial motion that Newton would later formalize in his first law of motion. The Merton Thinkers also made the first known statement of the 'mean speed theorem.'

Theorem. If a particle is uniformly accelerated from rest to some velocity, it will travel half the distance it would have traveled over the same interval with the final velocity.

For centuries, Galileo was thought to have been the first to state such ideas (compare his falling body discussion, pg. 100), but the Oxford group beat him by 250 years. They had no algebra with which to prove their claims, and essentially only asserted examples. For instance, if time intervals are in the proportion 2 : 1 and if particles subject to the same uniform acceleration are accelerated from rest over these intervals, then the resulting distances traveled will be in the proportion 4 : 1. In modern notation:

$$v_1 = at, \quad v_2 = 2at \implies d_1 = \frac{1}{2}v_1t = \frac{1}{2}at^2, \quad d_2 = \frac{1}{2}v_2 \cdot 2t = 2at^2 = 4d_1$$

In the 1350s, Nicolas Oresme (Paris) considered velocity geometrically by drawing what are essentially velocity-time graphs. This is roughly the approach taken later by Galileo, and another early version of *axes*. A major difference is that Galileo married mathematics to *observation*; uniform acceleration for Galileo was precisely the motion of a falling body.

A rigorous definition of *instantaneous velocity* is difficult because it requires *limits*: one measures average velocity over smaller and smaller intervals. You are in good company if you find this challenging: Zeno's arrow paradox is an objection to the very idea of instantaneous velocity! Even if one accepts the concept, its *direct* measurement, even in modern times, is nigh-impossible: for instance, radar Doppler-shift (as used to catch speeding motorists) requires measuring the wavelength of a radar beam, which amounts to computing the average velocity over a very small time interval.

The Area Problem pre-1600

We've previously seen two situations in which calculus-like methods described areas.

- Archimedes (sec. 3.4) computed the area inside a parabola using infinitely many triangles in what amounts to the evaluation of an infinite series. His 'cross-section' approach to finding areas and volumes also feels modern, though this work was unknown until 1899.
- Kepler (pg. 98) argued for his second law (equal areas in equal times) using infinitesimally small triangles to approximate segments of an ellipse: this is more akin to integration. He also applied this method to other problems, crediting Archimedes with the approach.

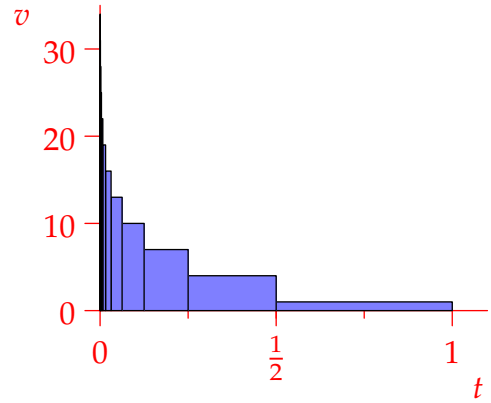
The modern integral (via Riemann sums) is a special case of approximating area using small rectangles: the philosophical challenge is again the notion of *limits* and infinitesimals.

In an early antecedent, Oresme describes how to compute the distance traveled by a particle whose speed is constant on a sequence of intervals. For example:

Over the time interval $\left[\frac{1}{2^{n+1}}, \frac{1}{2^n}\right)$ a particle travels at speed $1 + 3n$. How far does it travel in 1 second?

Oresme drew boxes to compute areas. In modern language, the distance traveled is an infinite series

$$d = \sum_{n=0}^{\infty} (1 + 3n) \left(\frac{1}{2^n} - \frac{1}{2^{n+1}} \right) = \sum_{n=0}^{\infty} \frac{1 + 3n}{2^{n+1}} = 4$$



Similarly to Archimedes, the infinite sum was evaluated by spotting two patterns:

$$\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} \cdots + \frac{1}{2^{n+1}} = 1 - \frac{1}{2^{n+1}} \quad \quad \frac{0}{2} + \frac{1}{2^2} + \frac{2}{2^3} + \frac{3}{2^4} + \cdots + \frac{n}{2^{n+1}} = 1 - \frac{n+2}{2^{n+1}}$$

Oresme had neither our notation nor our (limit-dependent) concept of an infinite series. He also worked with similar problems for uniform acceleration over intervals. These are not true Riemann sums, nor are they physical, for a particle cannot suddenly change speed.

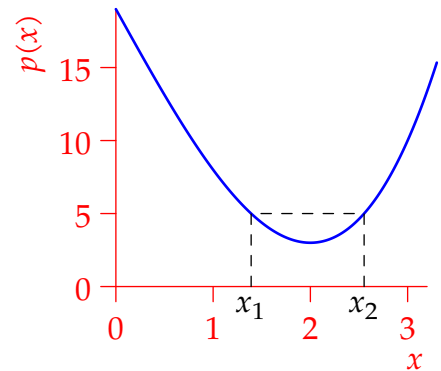
Calculus à la Fermat & Descartes

The advent of analytic geometry allowed Fermat and Descartes to turn the computation of instantaneous velocity and related differentiation problems into algebraic processes. The velocity of an object is identified with the slope of the displacement-time graph, which may be computed using variations of the modern method of *secant lines*. We discuss their competing methods.

Fermat's method of adequation We illustrate with an example. Our goal is to find the local minimum (located at $m = 2$) of the cubic function $p(x) = x^3 - 12x + 19$.

Fermat argues that if x_1 and x_2 are located near m such that $p(x_1) = p(x_2)$, then the polynomial $p(x_2) - p(x_1)$ (which equals zero!) is divisible by $x_2 - x_1$. Indeed

$$\begin{aligned} 0 &= \frac{p(x_2) - p(x_1)}{x_2 - x_1} = \frac{x_2^3 - 12x_2 + 19 - x_1^3 + 12x_1 - 19}{x_2 - x_1} \\ &= \frac{(x_2 - x_1)(x_2^2 + x_1x_2 + x_1^2 - 12)}{x_2 - x_1} \\ &= x_2^2 + x_1x_2 + x_1^2 - 12 \end{aligned}$$



Since this algebraic identity holds for any x_1, x_2 with $p(x_1) = p(x_2)$, Fermat claims that it holds also when $x_1 = x_2 = m$: note the assumption of continuity and limit-taking! He concludes

$$3m^2 - 12 = 0 \implies m = 2$$

By considering values of x near m , it is clear to Fermat that he really has found a local minimum. We recognize the idea that the slope of the tangent line is zero at local extrema.

Fermat's approach dates from the 1620s and is similar to earlier work of Viète. He later alters his method by considering values $p(x)$ and $p(x + e)$ for small e (x is 'adequated by e '), with the difference more easily divided by e without factorizing. Compare with the previous calculation:

$$\begin{aligned} 0 &= \frac{p(x + e) - p(x)}{e} = \frac{x^3 + 3x^2e + 3xe^2 + e^3 - 12x - 12e + 19 - x^3 + 12x - 19}{e} \\ &= \frac{3x^2e + 3xe^2 + e^3 - 12e}{e} = 3x^2 - 12 + 3xe + e^2 \end{aligned}$$

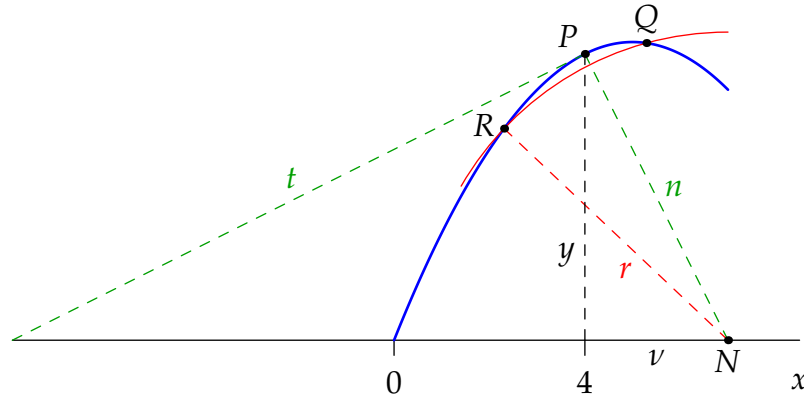
Finish by setting $e = 0$ and solving for x as before. Observe the derivative $p'(x) = 3x^2 - 12$ and how e plays the role of h in the modern definition

$$p'(x) = \lim_{h \rightarrow 0} \frac{p(x + h) - p(x)}{h}$$

Fermat's method works for any polynomial: in such cases, the limit definition of derivative requires no more than simple evaluation at $h = 0$. He moreover extended his method to cover implicit curves and their tangents.

Descartes' method of normals Descartes and Fermat are known to have corresponded regarding their methods. Descartes indeed seems to have felt somewhat challenged by Fermat, and engaged in some criticism of his approach. Descartes' method (in *La Géométrie*) relies on circles and repeated roots of polynomials in order to compute tangents.

In this example, we compute the slope of the parabola $y = \frac{1}{4}(10x - x^2)$ at the point $P = (4, 6)$.



Let $N = (4 + \nu, 0)$ be where the **normal** to the curve intersects the x -axis.⁶⁶ Draw a **circular arc** with radius r centered at N . If r is close to n , this circle intersects the curve in two points Q, R near to P . The line $\overleftrightarrow{QR}$ plainly approximates the tangent at P .

The co-ordinates of Q, R may be found using algebra: substituting $y = \frac{1}{4}(10x - x^2)$ into the equation for the circle results in an equation with two known roots, namely the x -values of Q and R . By the factor theorem,

$$\begin{cases} (x - (4 + \nu))^2 + y^2 = r^2 \\ y = \frac{1}{4}(10x - x^2) \end{cases} \implies (x - Q_x)(x - R_x)f(x) = 0$$

⁶⁶At the time, ν was known as the *subnormal* and t the *tangent*.

where $f(x)$ is some polynomial (in this case quadratic). Rather than doing this explicitly, Descartes observes that if the radius r is adjusted until it *equals* n , then $Q = R = P$ and the above equation has a *double-root*:

$$\begin{cases} (x - (4 + \nu))^2 + y^2 = n^2 \\ y = \frac{1}{4}(10x - x^2) \end{cases} \implies (x - P_x)^2 f(x) = (x - 4)^2 f(x) = 0$$

Factorization can be done by hand using long-division (note that ν and n are currently unknown!): substituting as above, we obtain

$$\begin{aligned} 0 &= x^4 - 20x^3 + 116x^2 - 32(4 + \nu)x + 16(4 + \nu)^2 - 16n^2 \\ &= (x - 4)^2(x^2 - 12x + 4) + 32(3 - \nu)x + 16(12 + 8\nu + \nu^2 - n^2) \end{aligned}$$

Since the remainder $32(3 - \nu)x + 16(12 + 8\nu + \nu^2 - n^2)$ must be the zero polynomial, we conclude that $\nu = 3$. By similar triangles, the slope of the curve at P is therefore

$$\frac{y}{\sqrt{t^2 - y^2}} = \frac{\nu}{y} = \frac{1}{2}$$

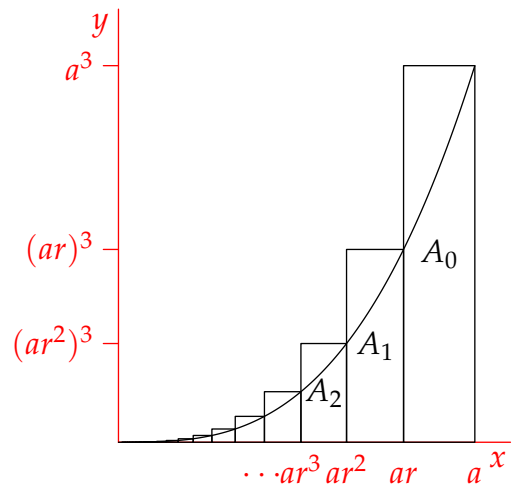
Fermat and Area The previous methods permit *differentiation*, albeit inefficiently. Fermat also approached the area problem in a manner similar to Oresme. Here is an example of this approach, where we find the area under the curve $y = x^3$ between $x = 0$ and $x = a$.

Let $0 < r < 1$ be constant. The rectangle on the interval $[ar^{n+1}, ar^n]$ touching the curve at its upper right-corner has area

$$A_n = (ar^n - ar^{n+1}) \cdot (ar^n)^3 = a^4(1 - r)r^{4n}$$

The sum of the areas is therefore

$$\begin{aligned} \sum_{n=0}^{\infty} A_n &= a^4(1 - r) \sum_{n=0}^{\infty} r^{4n} = \frac{a^4(1 - r)}{1 - r^4} \\ &= \frac{a^4}{1 + r + r^2 + r^3} \end{aligned}$$



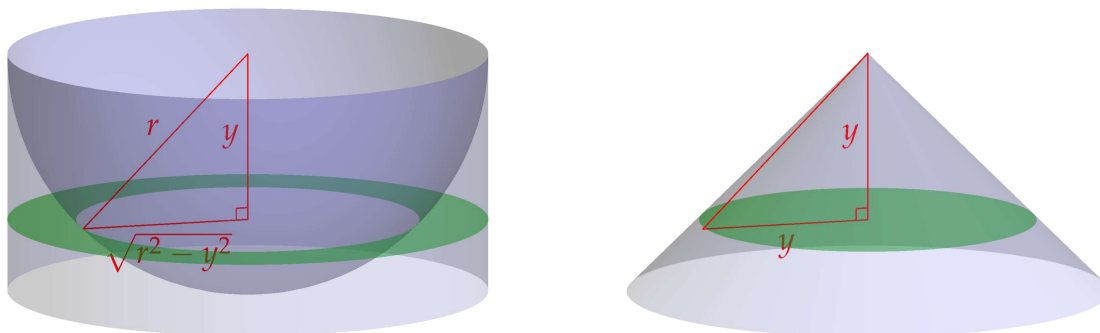
Setting $r = 1$ recovers the area under the curve: $\frac{1}{4}a^4$.

This is non-rigorous by modern standards and again implicitly invokes limits⁶⁷ by setting $r = 1$. By similar calculations, Fermat is able to establish the power law $\int_0^a x^n dx = \frac{1}{n+1}a^{n+1}$ for any positive integer n .

⁶⁷For Fermat, $r = \frac{n}{m}$ would have been rational, and he'd have set $m = n$ at the end as in his adequation method

Italian Calculus in the 17th Century: Area and Volume Problems

In contrast to the work of Fermat and Descartes, contemporary Italian scholars were more focused on integration problems. Here is Galileo's classic 'soup bowl' problem, where he compares the volume between a hemisphere and a cylinder to that of a cone.



Galileo observed⁶⁸ that the **cross-sectional areas** on both sides are equal (to πy^2). Since all cross-sections are equal, so must be the volumes. Unfortunately for Galileo, he couldn't sufficiently address two philosophical objections:

The zero-measure problem If cross-sections are 'equal,' then the top cross-sections state that a circle 'equals' a point.

Infinitesimals sum to the whole? Can we claim that equal cross-sections imply equal volumes?

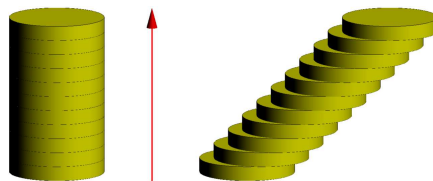
It was Galileo's advocacy on these points that first gained him notoriety with the Church. His later evangelism for the Copernican theory merely rekindled old animosities.

Bonaventura Cavalieri (1598–1647) Cavalieri, a student of Galileo and a Jesuit scholar, gave a more thorough discussion of indivisibles in 1635. He is particularly remembered for *Cavalieri's principle*:

If geometric figures have proportional cross-sectional measure at every point relative to a line, then the figures have measure in the same proportion.

Galileo's soup bowl is an example of this reasoning, where the 'line' is any vertical.

Another classic example involves sliding a stack of coins or a deck of cards; the volume of the slanted coin stack equals that of the cylinder.



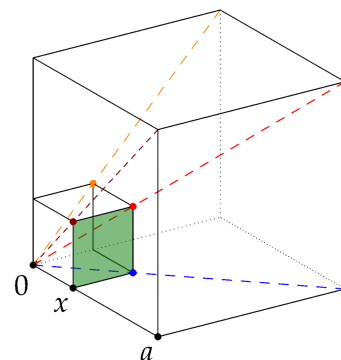
Extending his principle, Cavalieri inferred the power law $\int_0^a x^n dx = \frac{1}{n+1} a^{n+1}$, giving reasonable arguments for $n = 1$ and 2 . Here is a sketch of his approach for $n = 2$.

⁶⁸As did Archimedes 1900 years earlier (pg. 41), though Galileo was unaware of it.

Draw a cube of side x inside a cube of side a . This consists of three congruent pyramids with base a^2 and height a .

Consider the pyramid apex O and whose base is the square face nearest the viewer. The **cross-section** of this pyramid at position x has area x^2 . In Cavalieri's language, the pyramid is 'all the squares;' in modern language

$$\int_0^a x^2 dx = \frac{1}{3}a^3$$



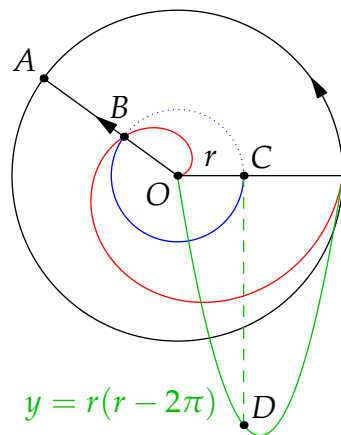
Cavalieri also used his method (Book IV, Prop 19 of *Geometria Indivisibilis*) to calculate the area enclosed in an **Archimedean spiral**. Suppose B moves at constant speed along a line $\overline{OA}$ rotating at constant speed about O . In polar co-ordinates B traces a curve $r = k\theta$ for $0 \leq \theta \leq 2\pi$ (we take $k = 1$).

If $|OB| = r$, then the **arc** inside the spiral has length

$$\ell(r) = 2\pi r \cdot \frac{2\pi - \theta}{2\pi} = r(2\pi - r)$$

Imagine the **arc** as a noodle which, when cut at B and allowed to fall, forms the **line** $\overline{CD}$. The area in the spiral therefore equals that within the parabola which (thanks to Archimedes and Cavalieri himself) is $\frac{4}{3}$ that of the largest triangle that can fit inside, namely

$$\frac{4}{3} \cdot \frac{1}{2}(2\pi)\ell(\pi) = \frac{4}{3}\pi^3$$

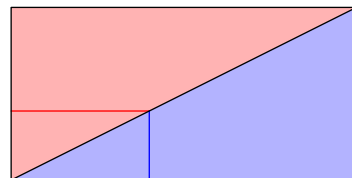


Cavalieri did this slightly differently, his parabola being drawn in a rectangle and the difference subtracted from a triangle, but the above is easier to visualize.

Cavalieri did not court controversy like Galileo, being well-aware of the contentious nature of indivisibles and taking great pains to distinguish 'all the lines/squares' of a figure from the figure itself. *Geometria Indivisibilis* was dense and difficult, so it wasn't easy to challenge. Cavalieri therefore remained relatively safe, even as his political rivals (the Jesuit order) worked hard to stamp out the 'dangerous' study of indivisibles.

Evangelista Torricelli (1608–1647) A contemporary of Galileo and Cavalieri, Torricelli made several applications of Cavalieri's principle and advocated for its careful use.

If the sides of a rectangle are in the ratio 2 : 1, so also are indicated red and blue segments. In Cavalieri's language, 'all the lines' of the red triangle are twice 'all the lines' of the blue triangle! Torricelli observes that the red triangle cannot have twice the area of the blue, since they are congruent and points out that Cavalieri's principle has been misapplied: the cross-sections were not measured with reference to a common line.



In modern language, $\int_0^2 \frac{1}{2}x \, dx = \int_0^1 2y \, dy$ are equal via the substitution $x = 2y$. The point is that the infinitesimals are also in the *same ratio* $dx : dy = 2 : 1$.

Another of Torricelli's examples offers a seeming paradox.

The hyperbola with equation $z = \frac{1}{x}$ is rotated around the z -axis. A **cylinder** centered on the z -axis with radius x lying under the surface has surface area

$$A = \text{circumference} \cdot \text{height} = 2\pi xz = 2\pi$$

Underneath the graph at x , Torricelli draws a **circular disk** with area 2π . Since the area of this disk is independent of x , he argues that the volume under the original **surface** out to radius a equals the volume of the solid **cylinder**:

$$V = 2\pi a$$

Torricelli argues that this is a correct use of Cavalieri's principle since the cylindrical 'cross-sections' and the circular cross-sections are both measured with respect to the same line (the x -axis).

This is precisely the method of volume by cylindrical shells that we learn in modern calculus:

$$V = \int_0^a 2\pi x \cdot \frac{1}{x} \, dx = 2\pi a$$

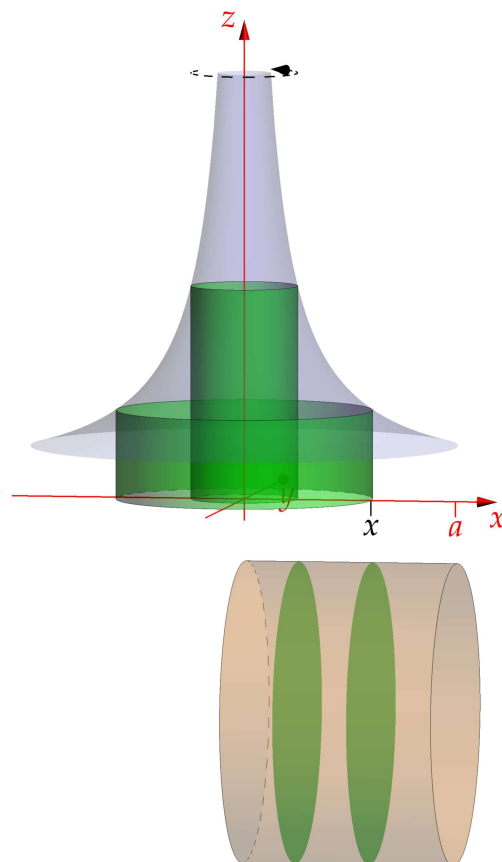
The conundrum is that the surface is *infinitely tall*! How can we justify the idea that it lies above a *finite* volume?

Galileo, Cavalieri and Torricelli mark the end of 400 years of Italian dominance of in science and mathematics dating back to Fibonacci. Their ideas were too controversial to thrive so close to the center of Church power. The center of European science and philosophy therefore moved northward. The English and French (protestant) reformations of the 1500s together with developing ideas of reformed government⁶⁹ meant that Northern Europe provided more fertile ground for new ideas.

Exercises 8.2. Key concepts:

1. Find the maximum of $p(x) = 5 + x - 2x^2$ using Fermat's first method.
2. Use Fermat's second method ("+"e") to find the maximum of $bx - x^3$. How might Fermat decide which of the two solutions to choose as his maximum?

⁶⁹For instance Hobbes' *Leviathan* written during the English Civil War (1642–1651) was a plea for the constraint of absolute monarchical power. The War itself indeed proved decapitatingly effective at reining in a King...

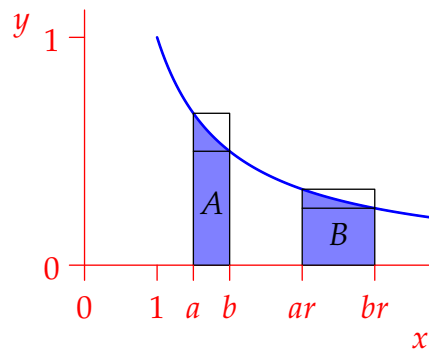


- Justify Fermat's first method of determining maxima and minima by showing that if M is a maximum of $p(x)$, then the polynomial $p(x) - M$ always has a factor $(x - a)^2$, where a is the value of x giving the maximum.
- Use Descartes' method of normals to compute the slope of the curve $y = x^2$ at (a, a^2) .
- Suppose the surface of a sphere of radius r is subdivided into infinitesimal regions of equal area. Following Kepler, use the formula for the volume of a cone ($\frac{1}{3}$ base·height) to find the relationship between the volume V of the sphere and its surface area A .
- (A problem of Kepler) Show that the largest circular cylinder that can be inscribed in a sphere is one in which the ratio of diameter to altitude is $\sqrt{2} : 1$.
(Hint: Relate the problem to finding the maximum of the function $f(x) = x - \frac{1}{4}x^3$, for which you can use modern calculus)
- We consider a version of Gregory St. Vincent's 1647 approach to the area under the hyperbola $xy = 1$.

- If $1 < a < b$ and $r > 0$ as in the picture, explain why the areas A and B under the curve satisfy the same inequalities

$$\frac{b-a}{b} < A, B < \frac{b-a}{a}$$

(Since $[a, b]$ may be subdivided into arbitrarily many subintervals, the areas A and B are therefore equal)



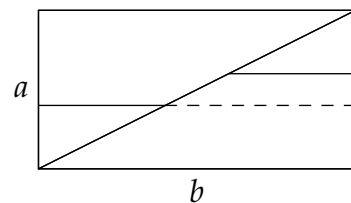
- If $A(x)$ is the area under the hyperbola between 1 and x , explain why A satisfies the logarithmic identity

$$A(x_1 x_2) = A(x_1) + A(x_2)$$

Why are you not surprised by this?

(For simplicity, assume $1 < x_1 < x_1 x_2$)

- Consider two copies of triangle with sides a, b arranged into a rectangle. Argue that 'all the lines' of the rectangle are twice 'all the lines' of the triangle.



(In modern language, $\int_0^a b \, dy = 2 \int_0^a \frac{b}{a} y \, dy$)

- Repeat Cavalieri's analysis of the spiral (pg. 114) to find the area inside one revolution of the curve $r = k\theta$ for any $k > 0$.